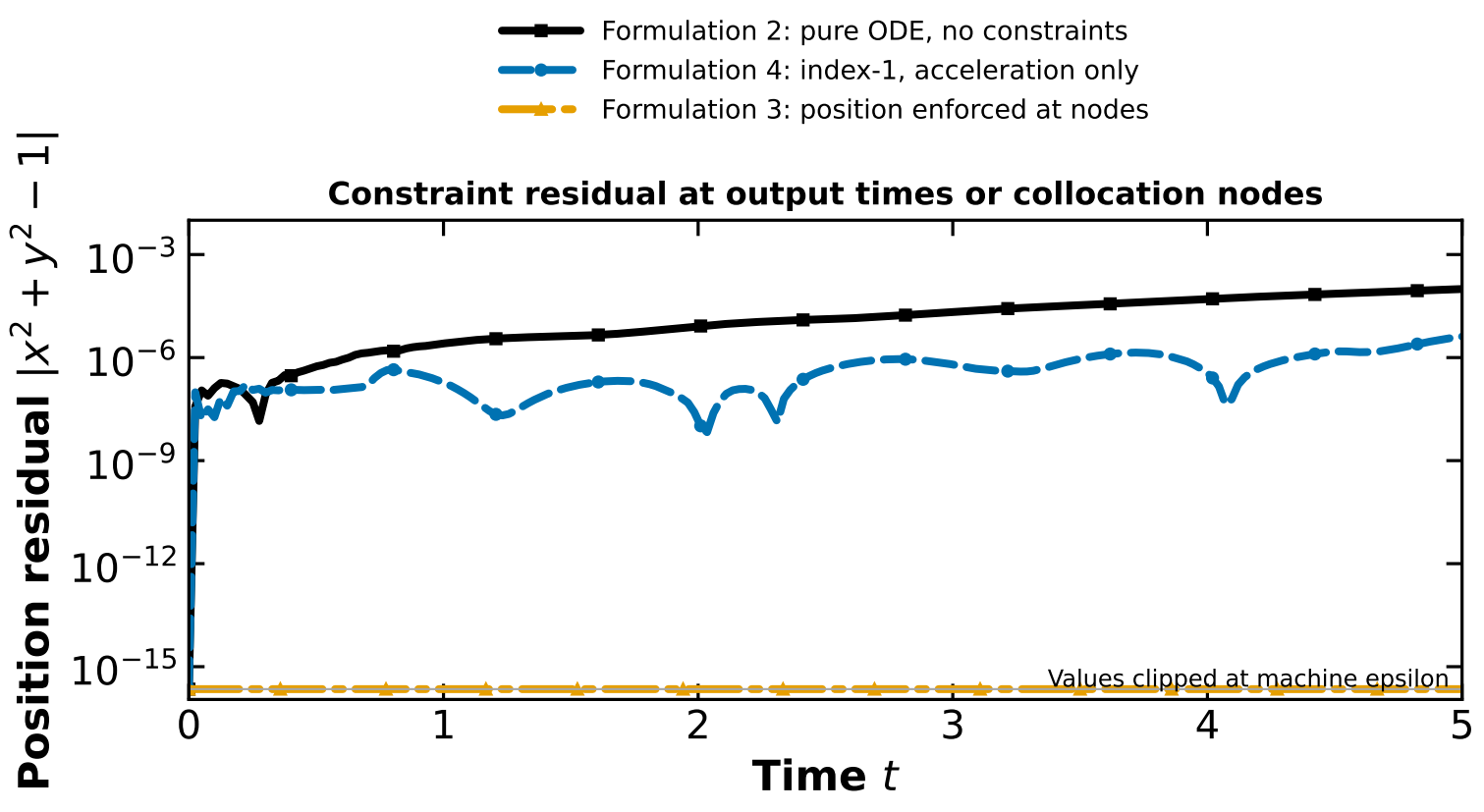
Formulation 2: pure ODE, no constraints
Formulation 4: index-1, acceleration only
Formulation 3: position enforced at nodes
Constraint residual at output times or collocation nodes
Position residual $|x^2 + y^2 - 1|$
$10^{-3}$
$10^{-6}$
$10^{-9}$
$10^{-12}$
$10^{-15}$
Values clipped at machine epsilon
0
1
2
3
4
5
Time $t$